

Analyzing Stock Market Volatility Patterns Using Historical Data

Anastasia S. Polskaya

Western Governors University

Table of Contents

A. Project Highlights	4
B. Project Execution	5
C. Data Collection Process	5
C.1 Advantages and Limitations of Data Set.....	6
D. Data Extraction and Preparation	6
E. Data Analysis Process	6
E.1 Data Analysis Methods	6
E.2 Advantages and Limitations of Tools and Techniques	7
E.3 Application of Analytical Methods	7
F Data Analysis Results.....	7
F.1 Statistical Significance	7
F.2 Practical Significance	8
F.3 Overall Success	8
G. Conclusion	9
G.1 Summary of Conclusions.....	9
G.2 Effective Storytelling.....	9
G.3 Recommended Courses of Action	9
H Panopto Presentation.....	9
References.....	12
Appendix A.....	12
Title of Appendix.....	13
Appendix B.....	13
Title of Appendix.....	14
Appendix C.....	14
Title of Appendix.....	15
Appendix D.....	15
Title of Appendix.....	16

A. Project Highlights

This project addressed the research question: *how can the analysis of historical stock prices over the past year help investors understand short-term volatility patterns in a selected stock?* To answer this question, the project analyzed data for Apple (AAPL) stock from September 2024 to September 2025. The scope of the project was narrow, focusing on the AAPL stock, daily observations for a period of 1 year, and the use of descriptive and diagnostic methods to calculate volatility and identify patterns that could inform short-term trading and portfolio management. Understanding volatility is essential for investors because it influences risk assessment and portfolio management. By analyzing historical data, this project's objective was to identify volatility trends that could help investors make informed decisions in the short term.

This project's scope included gathering one year of historical stock price data for the Apple (AAPL) stock, calculating volatility measures, completing descriptive and diagnostic analysis, and creating visualizations. The focus was on short-term volatility patterns such as daily returns, rolling volatility, and weekday patterns. The data used is publicly available, and it is historical data. Due to the focus being on historical data, this project's scope did not include predictions of future prices or volatility. Predictive analysis was outside of the scope.

The project's solution explored the historical daily stock price data for Apple Inc. to identify short-term volatility patterns. The solution calculated daily returns, volatility measures, rolling volatility, and weekday volatility patterns to assess behavior. A one-sample t-test was done to evaluate whether the average daily return is significantly different from zero. The results were shown through visualizations and statistical results that allow investors to interpret

volatility trends and assess short-term investment risks. The solution deliverables include a documented Jupyter Notebook containing the analysis and visualizations, as well as this written final report.

The methodology used in this project's solution was CRISP-DM. This methodology provided a structural framework for the project. Here is an overview of how the methodology organized my analysis:

- Business Understanding: This step is where I defined the research question(s) for the project.
- Data Understanding: In this step, I explored the Yahoo Finance data for AAPL stock and made sure that the needed fields are present.
- Data Preparation: This step is where I cleaned and transformed the dataset by checking for missing values and calculating any additional fields needed.
- Modeling/Analysis: Here I calculated volatility, created visualizations, and analyzed patterns.
- Evaluation: In this step, I interpreted the trends and patterns, making sure the research question got answered.
- Deployment/Reporting: In this step, I ensured that I had a complete Jupyter Notebook with documentation of findings and visualizations, and a final report.

This project was conducted in Jupyter Notebook, using Python. The main libraries that were used are: Pandas for data cleaning, preparation, and analysis, Matplotlib and Seaborn for data visualization, SciPy for the one-sample t-test, and yfinance for programmatically gathering historical stock data.

B. Project Execution

The project was executed with a few minor variances. The project plan was followed as originally intended: the project started with gathering a dataset of Apple's daily stock prices from Yahoo Finance using the yfinance library. The dataset was checked for completeness and cleanliness. Then, all values such as overall volatility and moving averages were calculated, all visualizations were created, and all code and discussion of findings were documented in a Jupyter Notebook. The only minor variance in this plan was that I forgot to mention the one-sample t-test in the plan as a deliverable/objective... This t-test was performed in the project to statistically evaluate whether the average daily return significantly differed from zero. It aligned with the project's objectives and strengthened the analysis without affecting the scope of the project.

The project planning methodology (CRISP-DM) was followed as planned; there were no variances because the methodology is straightforward and consists of logical steps that helped lead my project in phases. Each phase of the methodology was done in order. This methodology was effective because it provided a structured and logical approach for the project.

The project timeline and milestones slightly differed from my original plan. The data preparation, calculations, and visualizations were done as planned, but the written reports and task submissions took longer than planned. This delay was due to personal circumstances and the time required to thoroughly document everything. The analytical milestones were met as planned, and the variance impacted only the reporting phase rather than the execution of the analysis.

Milestone or deliverable	Duration (hours or days)	Projected start date	Anticipated end date	Actual End Date
Data preparation	3 days	10/8/2025	10/10/2025	10/10/2025
Calculations	3 days	10/11/2025	10/13/2025	10/13/2025
Visualizations	3 days	10/14/2025	10/16/2025	10/16/2025
Interpret Findings/Final Report	1 week	10/17/2025	10/24/2025	2/18/2026

C. Data Collection Process

The data was collected as planned, using the Python library ‘yfinance’ to extract historical daily stock prices for Apple Stock from Yahoo Finance. There were no unplanned issues or variances from the data collection plan, because I knew Yahoo Finance had the data that I needed, I knew it was a very reputable source, and I knew I wouldn’t have any issues collecting data from there. Although Yahoo Finance is reputable, I still validated that the data was accurate and complete. Also, the Python library ‘yfinance’ is quite reputable and popular. The data collection process can have obstacles in many cases, but in my case, I did not run into any obstacles. I was able to easily collect the data that I needed. In my data cleaning step, I ran tests to ensure cleanliness, and everything looked clean. There were no null values and nothing that looked out of place.

I did not run into any obstacles with the data collection and preparation step, because, as I already mentioned, Yahoo Finance is a very large and reputable source. It is held to a high

standard, which means it maintains its data well, and it is expected to be clean and accurate.

Although, no matter where data is from, it is always important to check it for cleanliness and completeness.

There were no unplanned governance issues either, because the data I collected and used from Yahoo Finance is publicly available financial data, and it does not contain any personally identifiable information. Standard precautions were taken to ensure proper data handling and documentation.

C.1 Advantages and Limitations of Data Set

An advantage of the dataset is its reliability and accessibility. Yahoo Finance is widely used and has regularly updated historical stock data. All of these factors make it suitable for financial analysis. The daily frequency of the data allowed for detailed analysis of short-term volatility patterns.

A disadvantage of the dataset is that it only includes one year of historical data, which limits the ability to analyze long-term volatility cycles or compare patterns across different market conditions. Also, because the data is historical and limited to a single stock, the findings may not generalize to broader market trends.

D. Data Extraction and Preparation

The data extraction process was performed programmatically using the `yfinance` library in Python, which allowed for efficient extraction of historical stock prices from Yahoo Finance. This was appropriate for data extraction because the `yfinance` library was specifically made to directly retrieve data from Yahoo Finance. Data preparation included cleaning the dataset, checking for missing or duplicate values, and calculating derived variables such as daily returns and volatility using Pandas. Pandas was used for the data preparation due to its ease of use and

efficiency in handling time-series financial data. This process was appropriate because it ensured a clean and complete dataset that was ready for analysis.

E. Data Analysis Process

E.1 Data Analysis Methods

This project used a few methods to analyze the data: descriptive analysis, diagnostic analysis, and statistical testing.

Descriptive analysis was performed to calculate financial measures, such as daily returns, volatility, rolling volatility, and moving averages. These are all standard measures used in financial analytics to identify patterns and assess risk. These measures summarized how Apple's stock price changed over time and provided a numerical understanding of price variability and risk. Descriptive methods were appropriate because they measure historical stock behavior and allow volatility trends to be observed, which helps increase understanding of stock behavior.

Diagnostic analytics was used to identify patterns and periods of increased risk. This included analyzing rolling volatility trends, comparing average volatility across weekdays, and using visualizations to easily see changes and patterns. Diagnostic methods were appropriate because they helped explain when volatility increased or decreased and showed patterns that could inform investment decisions.

A one-sample t-test was used to support the hypothesis, which was applied to Apple's daily stock returns. Daily returns were calculated as the percentage change between consecutive closing prices, and the one-sample t-test was used to compare the mean of these daily returns to a value of zero. This method allowed us to evaluate whether the average return differs from no return, providing an assessment of historical stock performance. A one-sample t-test was an

appropriate choice for supporting the hypothesis because it is designed to determine whether the average of a continuous numerical variable differs from a reference value. In financial analysis, zero is a common value for evaluating average stock returns, making this test relevant to the research question. The test aligned with the project goal of analyzing historical stock performance, rather than predicting future outcomes.

E.2 Advantages and Limitations of Tools and Techniques

Jupyter Notebook and Python were used to conduct and document the analysis. An advantage of Jupyter Notebook is that it provides a flexible and easy-to-use environment. It allows code, visualizations, and written explanations to be combined in one environment. A limitation of Jupyter Notebook is that because the outputs depend on execution history, reproducibility may be reduced if cells are not run in a consistent and organized order from start to finish. The Jupyter Notebook needs to be well-documented and organized. To address these limitations, the analysis was carefully documented, validation checks were done, and the notebook was executed in the correct order to ensure consistent results. An advantage of Python is that it integrates easily with libraries such as Pandas and Matplotlib that were used in the analysis. A limitation of Python is that it requires careful implementation and validation.

Pandas was used for data cleaning, preparation, and analysis. A limitation of Pandas is that it does not prevent logical data issues. Errors in data cleaning or manipulation can cause inaccuracies. This limitation was addressed by checking for missing and duplicate values and validating calculated columns such as daily returns.

Matplotlib and Seaborn were used for data visualization. An advantage of Matplotlib is that it allows detailed adjustment of plots, which makes it suitable for clearly presenting financial

trends and patterns. A limitation of Matplotlib is that it requires manual formatting to produce clear and interpretable visualizations. Poor axis labeling or formatting can lead to misleading graphs. This limitation was addressed by carefully labeling axes, using appropriate chart types for each variable, and visually reviewing plots to ensure that trends, volatility patterns, and comparisons are clearly seen. An advantage of Seaborn is that it simplifies the creation of statistically informative charts. A limitation of Seaborn is that it has less flexibility for customization, unlike Matplotlib. This limitation was addressed by using Matplotlib when more customization was needed.

SciPy was used to conduct the one-sample t-test statistical testing. An advantage of SciPy is that it provides reliable statistical functions that are widely used in analysis. A limitation of SciPy is that the statistical tests assume the data meet certain conditions, such as observations being independent and normally distributed. If these conditions are not met, the results may not be reliable. This limitation was addressed by ensuring that daily returns were continuous numerical data and visually inspecting return distributions.

E.3 Application of Analytical Methods

The data analysis followed a structured, step-by-step process. First, historical daily stock price data for Apple (AAPL) for the period of September 2024-September 2025 was gathered using the Python yfinance library. The dataset was assessed to confirm that the necessary fields were there, such as daily closing prices. Data preparation included checking for missing or duplicate values. Daily returns were then calculated using the percentage change between consecutive daily closing prices. This was necessary to convert raw price data into return values, which were needed for volatility calculations and statistical testing. The resulting daily return column was checked to make sure it consisted of continuous numerical values that were suitable for statistical

analysis. This calculation produced a missing (nan) value for the first observation row because there was no prior trading day for comparison. This missing value was expected, And was excluded from the analysis by dropping null values before conducting the one-sample t-test to ensure valid results.

Descriptive analysis was applied next. Summary statistics were calculated. Daily volatility was calculated using the standard deviation of daily returns, and rolling volatility was calculated using a rolling window to identify periods of increased market risk. Moving averages were calculated to smooth price fluctuations. This method required continuous numerical data, which was satisfied by the daily return and price variables.

Diagnostic analysis was done next. Visualizations, including line charts, volatility plots, return distributions, and weekday volatility, were created to identify patterns, periods of high volatility, and to explore differences across trading days. This required consistent time-ordered data and complete observations, which were verified during data preparation.

To test the project's hypothesis, a one-sample t-test was applied to the daily returns data. The null hypothesis stated that the mean daily return was equal to zero, while the alternative hypothesis stated that the mean daily return differed from zero. The t-test was conducted using SciPy's "ttest_1samp" function with a population mean of zero. The test generated a t-statistic and a p-value, which were compared to an alpha level of 0.05, to determine statistical significance.

The assumptions of the one-sample t-test were evaluated prior to interpreting the results. The daily return variable was continuous and numerical, satisfying the measurement requirement. The dataset combined a sufficiently large number of observations over a period of one year. Visual inspection of the return distributions and time-series plots was used to confirm

the absence of extreme inconsistencies that would invalidate the test. Independence of observations was reasonably assumed because each daily return represented a separate trading day. Based on these considerations, the data satisfied the assumptions necessary to apply the one-sample t-test.

F Data Analysis Results

F.1 Statistical Significance

The statistical analysis evaluated whether Apple's average daily return over the one-year period differed significantly from zero. The null hypothesis of the statistical test stated that the mean daily return of Apple's stock over the one-year period is equal to zero, indicating no average gain or loss. The alternative hypothesis stated that the mean daily return was not equal to zero.

A one-sample t-test was conducted using SciPy with an alpha level of 0.05. The test produced a t-statistic of 0.40 and a p-value of 0.69. Because the p-value exceeded the alpha level of 0.05, the null hypothesis was not rejected. This indicates that there was no statistically significant evidence that Apple's average daily return differed from zero during the period. These results suggest that while daily returns fluctuated over time, the average return across the one-year period was not statistically different from no return.

F.2 Practical Significance

Even though the one-sample t-test did not identify statistically significant average daily returns, the solution provided meaningful insights into Apple stock's short-term volatility behavior. The practical significance was demonstrated through the identification of periods of high volatility and fluctuations in daily returns over time. These periods of high volatility indicate increased market risk, which is important for short-term investment decision-making. In

financial decision-making, understanding the variability and timing of returns is often more important than identifying a statistically significant average return, especially for short-term risk management.

The descriptive and diagnostic analyses revealed periods of high volatility and variation in daily return behavior across the one-year timeframe. Rolling volatility measures showed areas of increased risk, which are practically significant because they indicate times when price fluctuations were larger, and uncertainty was higher. These periods of high volatility can help inform trading strategies, such as setting tighter stop-loss limits or temporarily reducing exposure to the stock. Additionally, the weekday volatility analysis showed differences in average volatility, suggesting that market behavior may vary based on the weekday. An investor may use this insight to plan entry and exit times and strategies on days with historically lower volatility.

A practical application of these findings would be for an investor to monitor rolling volatility indicators as an early warning sign for possible increased risk and therefore, adjust strategies accordingly. For example, during periods of anticipated increased volatility, an investor may choose to use a more cautious approach to manage risk. Even without statistically significant average returns, the insights gained from volatility patterns and the ability to anticipate volatile periods support informed decision-making by emphasizing risk awareness and informing decisions. Overall, the project demonstrates practical significance by improving understanding of short-term risk behavior, which directly supports more informed and responsible investment decisions.

F.3 Overall Success

Overall, this data analytics project was successful and effective in meeting its objectives and addressing the research question. The project successfully collected, cleaned, and analyzed one year of historical Apple (AAPL) stock price data and applied descriptive, diagnostic, and statistical methods to examine volatility trends. All planned calculations were completed accurately, such as the calculation of daily returns and volatility. Although the one-sample t-test showed that Apple's average daily return was not statistically different from zero at the 0.05 significance level, the result still represents a successful application of statistical analysis.

The project delivered meaningful insights by identifying periods of high volatility and patterns in short-term stock price behavior. The descriptive analysis calculated daily returns and volatility, while diagnostic analysis and visualizations clearly identified periods of increased risk and patterns in stock volatility behavior. The findings improved understanding of short-term stock volatility and demonstrated how historical data can support informed risk management and investment decision-making.

The project met all defined success criteria, including a clean dataset, accurate calculations, clear visualizations, a fully documented Jupyter Notebook, and a final report. Overall, the project met the defined success criteria, followed an appropriate analytical methodology, and produced useful results.

G. Conclusion

G.1 Summary of Conclusions

The analysis concluded that while Apple's average daily stock return over the one-year period did not differ significantly from zero based on the one-sample t-test, meaningful volatility patterns were still present in the data. Descriptive and diagnostic analysis revealed fluctuations in

daily returns, periods of increased rolling volatility, and variation in volatility across weekdays. These findings suggest that short-term risk levels changed over time even when average returns remained stable. The results support the conclusion that historical stock price analysis is valuable for understanding short-term volatility behavior, even when tests do not indicate statistically significant results. significant average returns. Overall, the project demonstrated that volatility analysis can provide important insights into market behavior that are useful for informing investor decision-making.

G.2 Effective Storytelling

The tools used and the visualizations effectively supported storytelling by making volatility patterns easier to identify. Line charts showing closing prices, rolling volatility, daily returns, and moving averages allowed trends and periods of heightened risk over time to be seen visually. Histograms of daily returns showed the distribution and variability of returns, helping to interpret the results of the statistical test. Bar charts comparing average volatility across weekdays highlighted differences that may not have been noticed from summary statistics. These visualizations, created using Matplotlib and Seaborn Python libraries within the Jupyter Notebook, complemented the numerical analysis and allowed findings to be communicated clearly. Together, the graphical representations and written analysis provided a cohesive narrative that connected the research question, methods, and results.

G.3 Recommended Courses of Action

Based on the results of this analysis, two recommendations can be suggested. The first one is that investors can use short-term volatility metrics, such as rolling volatility and daily returns, to inform risk management strategies rather than solely relying on average returns.

Monitoring times of increased volatility may help investors adjust their strategies, such as timing, to avoid increased risk. Second, organizations or analysts can expand by applying the same analytical framework to multiple stocks or longer time periods to compare volatility behavior. This would allow for broader insights into an investor's portfolio and help further inform their decision-making and strategies. Both recommendations support the research question by showing how historical stock price analysis can improve understanding of short-term volatility patterns.

H Panopto Presentation

<https://wgu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=83d8c623-0801-418b-a504-b3f600ae47df>

References

Apple Inc. (AAPL) stock historical prices & data - yahoo finance. Yahoo Finance. (n.d.).
<https://finance.yahoo.com/quote/AAPL/history/>

No sources were cited.